Jyothiradithya Lingam

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Education

Rajiv Gandhi University of Knowledge Technologies, IIIT Nuzvid Jul 2025 – May 2029
B.Tech in Artificial Intelligence and Machine Learning **CGPA: 9.1/10**

Rajiv Gandhi University of Knowledge Technologies, IIIT Nuzvid 2023 – 2025
Pre-University Course (PUC) **CGPA: 9.82/10**

Research Competitions

3rd Place – GOOSE 2D Fine-Grained Semantic Segmentation Challenge Mar 2026 – May 2026
ICRA 2026 | First author, 3-member team arXiv:2606.15937 | Code

- Developed **GOOSE-M2F**, a long-tail-aware Mask2Former (Swin-L) for 64-class outdoor terrain segmentation, adding an ASPP + CBAM refinement module, a high-resolution auxiliary head, and 200 object queries.
- Tackled class imbalance with distribution-balanced loss, rare-class copy-paste, and class-aware repeat sampling.
- Achieved **70.08% official composite mIoU**; designed a dense sliding-window inference pipeline with Gaussian blending and multi-scale flip TTA that raised the score from 59.51%.

16th Place – MoCha 2026: Benchmark and Challenge on Parkinsonian Gait May 2026 – Aug 2026
ECCV 2026 | 1,669 submissions, 113 participants, Code

- Contributed to a cross-site Parkinsonian gait severity prediction pipeline fusing **BiGRU/MoMask** motion embeddings with engineered biomechanical features.
- Applied adversarial feature selection and calibrated ensemble learning to generalize across recording sites.

18th Place – Universal UltraSound Image and Video Analysis Challenge 2026 Jun 2026 – Aug 2026
Deep-Breath 2026 at MICCAI | 1,137 submissions, 99 participants Code

- Developed a unified multi-branch framework for multi-organ ultrasound segmentation and diagnostic classification across B-mode images, CEUS clips, and echocardiography video.
- Combined **EfficientNet-B5 + UNet++** segmentation trained with a Dice–Focal–BCE loss, 16-frame temporal pooling for CEUS perfusion classification, and morphological refinement; packaged as a Docker submission.

33rd Place – UPAR Challenge, Track 2: Attribute-Based Person Retrieval Sep 2026 – Oct 2026
RWS 2027 at WACV 2027 | 695 total submissions, 79 participants Code

- Developed a **cross-domain person retrieval** framework for real-world surveillance datasets, combining visual representations with attribute-aware learning.
- Applied domain-shift calibration and cascaded re-ranking for fine-grained retrieval across datasets.

Hackathon Projects

MacBook Pro Award – Build for Good Hackathon 2026 August 2026
Satyamev-Bot | Top 34 of 3,000+ teams nationwide Code

- Built **Satyamev-Bot**, a WhatsApp-based agentic RAG fact-checker that verifies text, screenshot, and voice-note claims in 10 Indian languages plus English, returning a verdict, confidence score, and cited sources.
- Integrated Whisper ASR and Gemini Vision OCR with a LangChain agent using real-time web search; served via FastAPI and Twilio, containerized with Docker, and auto-deployed to Hugging Face Spaces with GitHub Actions.

Top 10 Finalist – Megathon 2026, IIIT Hyderabad October 2026
SvaraAI | Accessible AI Track Code

- Built **SvaraAI**, an Android app (Kotlin, Jetpack Compose) that helps visually, hearing-, and speech-impaired diners read menus, check dietary restrictions, and order through speech-to-speech interaction in 6 languages.
- Integrated **Bhashini** ASR, translation, and TTS with ML Kit OCR and Llama 3.3 70B for menu and diet reasoning.

Technical Skills

- **Programming:** Python, Java, C
- **Areas:** Computer Vision, Deep Learning, Machine Learning, AI Engineering, LLMs, VLMs
- **Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, Hugging Face, timm, CUDA, LangChain, FastAPI
- **Tools:** Git, GitHub Actions, Docker, VS Code, Kaggle, Google Colab, TensorBoard, Weights & Biases

Research Interests

Computer Vision; Robust Deep Learning under Class Imbalance and Distribution Shift; Medical Image Analysis